

SHUBHAM KATHAVALE

COMPUTER SCIENCE ENGINEER & AI SYSTEM DEVELOPER

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EXECUTIVE SUMMARY

Obsessed with bridging the gap between hardware automations and intelligent AI systems. Extensive hands-on experience building complex, event-driven desktop companions (J.A.R.V.I.S. Personal AI OS) and full-stack web applications. Passionate about AI integration, software engineering, and creating automated workflows. Looking for an internship or junior developer role where I can build production-grade software solutions.

TECHNICAL SKILLS

Languages & DBs: Python, Java, JavaScript, HTML5, CSS3, SQL, MySQL, MongoDB, SQLite

AI & Automation: PySide6/Qt, RAG (FAISS), STT/TTS (Whisper), Subprocess, PyAutoGUI

Frameworks & Web: React.js, Node.js, Express.js, Firebase, Tailwind CSS

Domains: AI & Systems, Cloud Computing, Cyber Security, Software Engineering

PROJECTS

J.A.R.V.I.S. — Personal AI Operating System

Stack: PySide6, Qt, QWebEngine, Groq API, SQLite, FAISS Vector DB, Next.js, Zustand, Framer Motion

- Developed an advanced voice-activated desktop assistant transitioning into a proactive operating system companion.
- Designed an **Event-Driven Multi-Agent Architecture** coordinating sub-units (Memory, Knowledge, Task, Desktop, Vision).
- Implemented **Cognitive Memory** using SQLite permanent store to track user context, facts, goals, and preferences.
- Integrated **Knowledge Intelligence** using RAG with FAISS vector indexing on local project files and documentation.
- Built the **V.A.U.L.T. HUD Dashboard** running on a PySide6 QWebEngineView window at 60 FPS.

ARIA — Agentic Reasoning & Intelligence Assistant

Stack: Python, WebSockets, Ollama (Qwen2.5:7b), Claude API, Playwright, HTML5/CSS3, Split Canvas

- Designed and built an autonomous agentic execution loop with real-time WebSocket thought streaming and action logging.
- Implemented a custom schema adapter translating Claude-style tool definitions to OpenAI/Ollama-compliant schemas.
- Developed a 7-tool system execution suite (web_search, read/write file, run_python, run_shell, call_api, Playwright browse).
- Engineered a futuristic glassmorphic split Canvas UI featuring a multi-file workspace selector and code/preview modes.

SalonOS — Enterprise Salon Operating System

Stack: Python 3.13, FastAPI, SQLite/PostgreSQL, SQLAlchemy, Role-Based Access Control, WebSockets, HTML5, Vanilla CSS3, WebGL, Chart.js

- Engineered a multi-tenant operating system backend isolating database records per tenant using foreign keys.
- Built a responsive client booking widget featuring an interactive WebGL canvas, before/after sliders, and a 7-step reservation wizard.
- Developed an administrative dashboard displaying today's business KPIs, polled live activity feeds, and channels simulator.
- Created a comprehensive integration test suite verifying JWT-based RBAC, stylist commissions, and checkout transaction logic.

Evokn Fitness — Full-Stack Fitness Tracker

Stack: React.js, Node.js, Firebase, Tailwind CSS

- Designed and built a responsive full-stack fitness application to track calorie intake, macros, and workout sessions.
- Configured secure **Firestore Authentication** and managed a real-time Firestore database for user analytics.
- Integrated dynamic analytics charts using dashboard libraries to track weight loss progression and nutrient goals.
- Successfully debugged complex database synchronization issues and optimized client-side state management.

Noisecity RP — GTA V FiveM Multiplayer Community

Stack: FiveM API, LUA, SQL, QBCore framework

- Orchestrated and developed custom scripts for a GTA V Roleplay (RP) server serving a growing community.
- Managed database linkages via MySQL to track player assets, character vitals, and economy transactions.

EDUCATION

B.Tech in Computer Science & Engineering (Direct 2nd Yr / Lateral Entry) <i>MIT World Peace University (MIT-WPU), Pune</i>	Expected 2028 <i>Pune, India</i>
Diploma in Computer Engineering <i>B.L. Patil Polytechnic MSBTE Board (First Class with Distinction)</i>	Completed 2024 <i>Maharashtra, India</i>

LICENSES, CERTIFICATIONS & COURSES

- **Oracle Cloud Infrastructure 2025 Certified Foundations Associate** (Issued Aug 2025 · Expires Aug 2035 | Credential ID: OC6817049)
- **University Course: Bioscience Using Python Programming (BTU10010)** — MIT-WPU
- **University Course: Business Analytics using Power BI (BBA20180)** — MIT-WPU